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EVOLUTIONARY BIOLOGY

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A synthesized framework for the evolutionary origins of avian obligate brood parasitism

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The repeated evolution of avian obligate brood parasitism (OBP) across diverse lineages represents a major evolutionary puzzle. While host-parasite coevolution has been extensively studied, the initial conditions favoring OBP emergence remain unclear. Here, we integrate theoretical modeling, paleoclimate mapping, and phylogenetic comparative analyses to identify the key drivers underlying OBP evolution. Our model shows that rising parental care costs, particularly under environmental stress, may favor the emergence of OBP. However, its persistence requires additional life-history adaptations, including increases in fecundity (egg production). Paleoclimatic data support this framework: OBP origins often followed major climatic upheavals with fixations occurring during subsequently more stable periods. Phylogenetic analyses reveal that OBP is associated with high-mortality ecological traits, such as open-habitat use. We also find increased OBP likelihood in both smaller carnivorous and larger herbivorous/omnivorous species, indicating size-dependent trophic constraints. These results provide a unified framework explaining the recurrent evolution of OBP by linking environmental pressures and life-history trade-offs to this remarkable behavioral adaptation.

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INTRODUCTION

Avian obligate brood parasitism (OBP) represents one of the most notable examples of evolutionary cheating in nature, where females abandon all parental care by depositing eggs exclusively in the nests of other species (*1*, *2*). This reproductive strategy allows parasitic parents to save energy by transferring the entire burden of offspring care to unwitting hosts (*3*). For host species, however, being parasitized should be inherently costly (*4*), especially when the parasites have evolved egg/chick eviction or killing behavior (*5*). In response, hosts have evolved various defensive strategies, including nest defense, egg rejection, chick discrimination, or fledgling abandonment (*6*).

The resulting coevolutionary arm races between parasites and hosts have been widely studied (*6*–*10*) and recognized as a powerful driver of trait evolution (*11*), phenotypic diversification (*12*), and dynamics of speciation and extinction (*13*–*15*). This rich body of research has illuminated the mechanisms that maintain OBP once established, yet how this remarkable strategy initially emerged and evolved remains one of the most compelling unsolved puzzles in evolutionary biology (*2*). Understanding its evolutionary process is crucial for developing a comprehensive framework that explains not only how OBP persists and diversifies but also why it evolves repeatedly across phylogenetically distant lineages.

Several hypotheses have been proposed for the evolutionary precursors to OBP, including nest-site takeover (*16*) and cooperative breeding system in which helpers may lay eggs in the nests they assist (*17*). These behaviors represent intermediate steps toward full parasitism. More fundamentally, OBP could evolve either directly from ancestral parental care (*18*) or through an intermediate stage of intraspecific brood parasitism (*19*–*21*). Evolutionary pathway analyses of cuckoo lineages support the former pathway (*1*, *22*), suggesting a sequence involving initial reductions in brain and body size, followed by the evolution of parasitism, and subsequently changes in life-history traits such as egg size reduction, transition to migratory behavior, and habitat shifts (*2*).

While these studies provide valuable insights into evolutionary sequences of OBP, they do not address the fundamental question of why OBP evolves in the first place. What ecological or evolutionary pressures initially drive the abandonment of parental care in favor of a seemly risky parasitic strategy, given that parasitic females must achieve higher fitness to make this transition viable? In addition, what factors subsequently promote the successful establishment and persistence of OBP lineages, where sustained higher fitness of parasitic females is likewise essential for maintaining these strategies over evolutionary time? Without answering these questions, we lack the theoretical foundation necessary to understand the repeated evolution of this reproductive strategy and to predict the conditions under which it might emerge.

To address these gaps, we integrated three complementary approaches to investigate the ecological and evolutionary origins of OBP. First, we constructed a simple optimization model comparing alternative reproductive strategies to identify the ecological conditions that favor a shift from parental care to parasitism. The model highlights the critical roles of elevated parental care costs and environmental instability as selective pressures that may favor the emergence of OBP, whereas the long-term maintenance and fixation of parasitic strategies likely depend on specific life-history adaptations, such as increased clutch sizes. We propose that favorable environmental conditions may create windows of opportunity for such evolutionary innovation.

Next, we investigated paleoclimate data to test whether OBP origins coincide with environmental transitions predicted by the model and conducted phylogenetic comparative analyses across the avian tree of life to identify specific life-history traits associated with OBP evolution. Together, these results provide a comprehensive framework linking environmental pressures, life-history constraints, and ecological opportunity in explaining the remarkable convergent evolution of OBP across diverse avian lineages.

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RESULTS

Optimization model

To explore the ecological conditions that could favor the evolution of OBP, we developed a simple optimization model following the framework of Lerch and Servedio (*23*). The model compares the lifetime reproductive success of individuals using different reproductive strategies. We hypothesize that shifts in key fitness-related ecological factors, such as rising parental care costs driven by increased predation risk, resource limitation, greater exposure to pathogens or parasites, or declining habitat quality, may trigger the evolutionary transition from parental care to parasitism.

In the model, we assume that individuals adopt a reproductive strategy defined by parameter p, representing the intention to conduct parasitism. Individuals reproduce via parental care with probability $1 - p$. A strategy of $p = 1$ corresponds to OBP, while $p = 0$ denotes pure parental care. Each individual reproduces once per breeding bout, either through parasitism or parental care, according to its strategy. Successful reproduction via brood parasitism depends on host response. We assume that parasitism is detected by the host with probability a and, if detected, is rejected with probability r. Thus, the overall probability of successful parasitism is $(1-a) + a(1-r) = 1 - ra$, accounting for both undetected attempts and detected-but-tolerated events.

Individuals may die between reproductive bouts at a baseline rate of d, which we model as a monotone decreasing function of environmental condition x, where $0 < x < 1$. Lower values of x correspond to harsher environments and are associated with higher mortality rates. Parental care typically incurs additional energetic costs (*3*), so we assume that individuals performing parental care have an added mortality risk c, which also depends on environmental conditions. For illustration, we use a linear function $c = k\,d(x)$ to model the additional cost of parental care, where $k > 0$ is a constant representing the relative intensity of parental care costs. Accordingly, the probability of an individual surviving to the next reproductive bout is given by $1 - [d + (1-p)c]$.

On the basis of these assumptions, we derive the expected lifetime reproductive fitness of individuals with a given strategy p and identify the fitness-maximizing value p^*. This value indicates whether pure parental care ($p^* = 0$), OBP ($p^* = 1$), or facultative brood parasitism (FBP; $0 < p^* < 1$) is the optimal strategy under the given environmental condition x. Full model details are provided in Materials and Methods.

Modeling results

Overall, we find that the brood parasitism strategy can be more adaptive than parental care when the costs associated with care increase (i.e., higher k; Fig. 1A). This is obvious given that parental care suffers additional mortality risks. This advantage of parasitism is further amplified under more challenging environmental conditions (i.e., lower x).

Specifically, when we model the death rate as a linear function of x ($d(x) = 1 - x$), brood parasitism is the optimal strategy when

$$k > \frac{x\,ra}{1 - ra} \tag{1}$$

parental care is favored when

$$k < \frac{-1 - ra(1-2x) + \sqrt{1 + r^2a^2 + 2ra(1-2x)}}{2ra(1-x)} \tag{2}$$

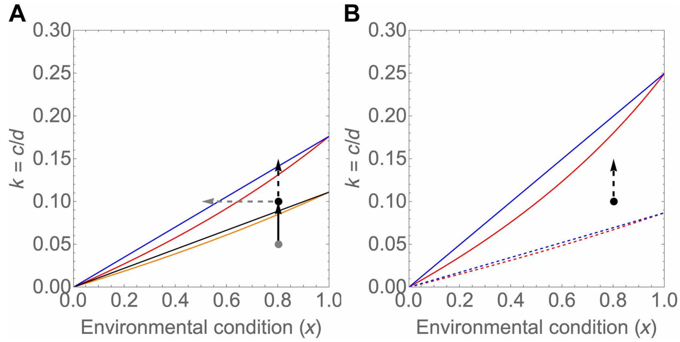

Fig. 1. Optimal strategy (p^*) under different environmental conditions (x) and parental care costs (k). (**A**) Effect of different host detection (a) and rejection rates (r). For $ra = 0.1$, OBP ($p^* = 1$) is optimal above the black line (i.e., when $k > \frac{x\,ra}{1-ra}$), and parental care ($p^* = 0$) is optimal below the orange line [i.e., when $k < \frac{-1-ra(1-2x)+\sqrt{1+r^2a^2+2ra(1-2x)}}{2ra(1-x)}$]. FBP ($0 < p^* < 1$) is favored when k falls between these two threshold lines. Increasing ra to 0.15 shifts the orange and black lines upward to the red and blue lines, respectively. For illustration, the lower gray point marks $x = 0.8$ and $k = 0.05$, where parental care is optimal. Brood parasitism will become optimal if parental care costs increase from 0.05 to 0.1 (black arrow). However, if the ra value increases from 0.1 to 0.15, parental care becomes optimal again at $x = 0.8$ and $k = 0.1$. The black and gray dashed arrows indicate scenarios where further increasing parental care costs or declining environmental conditions allow brood parasitism to once again outcompete parental care. (**B**) Effect of higher offspring production in parasites. When $ra = 0.2$, parental care is optimal at $x = 0.8$ and $k = 0.1$. However, if parasitic individuals produce more offspring than parental care individuals ($f_1 > f_2$), brood parasitism can persist as the optimal strategy. This shifts the threshold lines (blue and red lines) downward (blue and red dashed lines), placing the black point in the OBP-optimal region. A linear baseline death rate of $d(x) = 1 - x$ is considered here. In (B), $f_1 = 1.15$ and $f_2 = 1$ model unequal offspring production; otherwise, $f_1 = f_2 = 1$.

and FBP is optimal when parental care cost (k) lies between these two thresholds.

It can be seen that when host detection and rejection rates are low (i.e., when ra is small), even a modest increase in parental care cost (k) can make OBP the optimal strategy (black arrow in Fig. 1A). As hosts experience increased parasitic pressure, they may evolve stronger antiparasitic defenses, leading to higher detection and rejection rates (i.e., a higher value of ra). Under such conditions, a further increased cost of parental care is required to maintain parasitism's advantage (black dashed arrow in Fig. 1A). Furthermore, when environmental conditions become more challenging (i.e., decrease x), OBP may become favorable even at moderate parental care costs (see the gray dashed arrow in Fig. 1A). We also suggest that such challenging conditions could potentially amplify the energetic demands of parental care, further tipping the balance in favor of parasitism.

Life history and ecological conditions favoring OBP

As environmental conditions may later gradually improve, parental care can once again become the optimal strategy. For OBP to persist over evolutionary time, additional adaptive changes, such as laying more eggs (*24*), may be required. Our model shows that if parasitic individuals can evolve slightly higher offspring production, parasitism can remain optimal even under improved conditions (e.g., in Fig. 1B, the red and blue dashed lines lie below the black point, indicating that OBP is optimal). The evolved greater reproductive output could further elevate the burden of care for nonparasitic individuals (*25*), reinforcing the relative advantage of parasitism (see the black dashed arrow in Fig. 1B).

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To further explore the conditions favoring OBP evolution, we use the above theoretical framework to deduce the optimal strategy under three different scenarios of baseline death rates. The convex and concave functions are used to model the species that are more and less sensitive to environmental condition decline, respectively (see Materials and Methods). We find that OBP is more likely to evolve under the scenario with a convex function of $d(x)$, as brood parasitism can be optimal under a larger parameter range than that under the other two scenarios (see Fig. 2). It thus implies that brood parasitism should be more likely to evolve in those lineages that are more sensitive to environmental decline, such as species with reduced competitiveness for resources.

We also modified our model by setting individuals to reproduce only once (i.e., with a semelparous life history) (*23*). The modeling results clearly indicate that OBP is less likely to evolve under this scenario, unless parental care costs are very high or environmental conditions are particularly challenging (Fig. 2). However, increased fecundity (i.e., higher offspring production) can again favor the evolution of OBP across all death rate scenarios, even under a semelparous life history (fig. S1).

Our model can further be extended to consider scenarios where reproductive output varies with environmental conditions. If potential hosts are less affected by environmental conditions than their parasites (e.g., because of their superior competitiveness in securing resources), then the survival of parasitic offspring to breeding age would presumably be higher than that of self-raising offspring. We incorporated this factor into the model by assuming differential environment-dependent offspring survival rates, with $d_1 x$ and $d_2 x$ representing host- and self-raised offspring survival, respectively, where $d_1 > d_2$ and x is the environmental condition. This modification increased the likelihood of OBP evolution, as reflected by a downward shift of both threshold lines (fig. S2).

In summary, our modeling results suggest that a combination of elevated parental care costs and more challenging environments plays a pivotal role in the evolutionary origin of OBP. Furthermore, the long-term persistence and fixation of parasitism in OBP lineages

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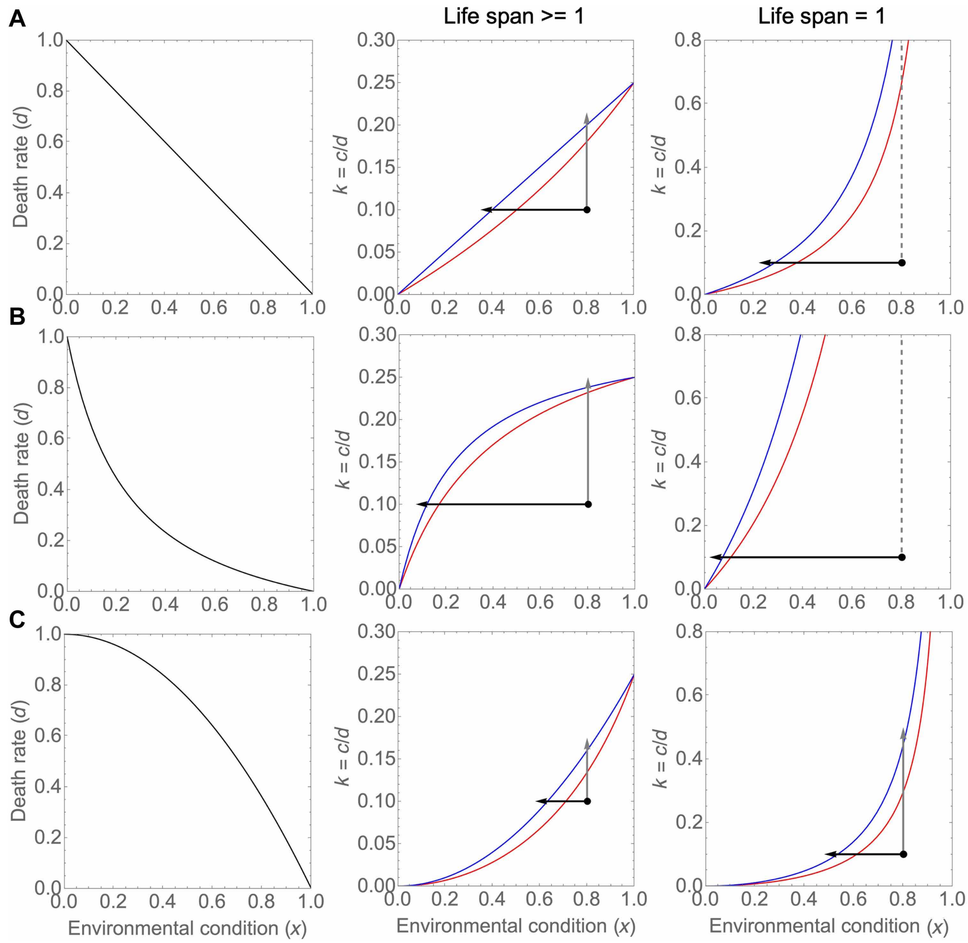

Fig. 2. Optimal strategy (p^*) under different baseline death rate functions $d(x)$. The blue and red lines represent the thresholds defined in Fig. 1. In (**A**), $d(x) = 1 - x$; in (**B**), $d(x) = \frac{1-x}{mx+1-x}$ is a concave function, indicating that individuals are relatively tolerant to the initial environmental deterioration; and in (**C**), $d(x) = 1 - x^{\alpha}\ (\alpha > 1)$ is a convex function, indicating that individuals are highly sensitive to environmental condition decline. Brood parasitism becomes optimal more easily under a convex $d(x)$ function, as indicated by the relatively shorter black and gray arrows in (C). In addition, when individuals reproduce only once (i.e., life span = 1), brood parasitism is less likely to be optimal compared to situations where life span >1 [see the relatively longer black arrows in the right-hand panels in (A) to (C)]. The gray dashed lines in the right panels of (A) and (B) indicate that even if parental care is quite costly, brood parasitism cannot become optimal in these cases. The black point represents a scenario where $x = 0.8$ and $k = 0.1$ for illustration. Other parameters are as follows: $f_1 = f_2 = 1$ and $ra = 0.2$.

may depend on the evolution of life-history traits such as increased fecundity, lower competitive ability, or iteroparity, which can reinforce the benefits of parasitism even after the ecological pressures that originally favored its emergence have lessened.

Theoretical predictions and the rarity of OBP

Interspecific OBP has evolved independently seven times among extant birds, representing ~1% of species (*26*, *27*), although the actual number of origins may be higher because of potential extinctions of parasitic lineages. Given the high costs of parental care, there is a long-standing puzzle as to why brood parasitism is not more prevalent (*28*). Our theoretical modeling provides insights into this paradox. As illustrated in Figs. 1 and 2, elevated parental care costs alone can confer a selective advantage to parasitic strategies but only when host detection and rejection rates remain low (i.e., with relatively small *ra* values). Challenging environmental conditions may further amplify this advantage, particularly as host selective pressure intensifies host antiparasitic defenses.

We suggest that the ancestral state of modern birds likely involved inherently costly parental care (e.g., nest building, incubation, and posthatch care of offspring), making OBP a potentially advantageous alternative strategy during its evolutionary origins. In the context of our model parameters, many ancestral bird species might inhabit the region of parameter space above the threshold lines where parasitism would be theoretically beneficial (see Fig. 1). However, brood parasitism rarely evolves to fixation because additional barriers may increase its effective costs. These include the difficulty in locating and accessing sufficient host nests during laying periods, uncertainty in securing effective heterospecific parental care, and developmental risks such as misimprinting on inappropriate host species. These factors functionally increase the costs of parasitism, similar to elevating *ra* values in our model (see the upward shifts of threshold lines in Fig. 1A), thereby constraining the evolution of obligate parasitism.

This raises a key question: What ecological conditions allow certain lineages to overcome these barriers and evolve OBP? We propose that abundant, syntopic host populations are critical for OBP evolution and persistence. Stable and favorable environmental conditions would facilitate the maintenance of high host population densities needed for parasitic success. Considering the time needed for organisms to attain and sustain the adaptive parasitic strategy (*29*) and the subsequential progress of speciation (*30*), the evolution and maintenance of OBP likely require initial ecological pressures that favor reduced parental investment, followed by sufficiently stable conditions to sustain the high-density host populations essential for parasitic success.

Paleoclimatic analysis of OBP origins

To test our theoretical predictions, we examined whether OBP origins coincided with periods of environmental instability. Although the precise timing of the behavioral shift to OBP is uncertain and may not align exactly with divergence dates of the parasitic clades, it most likely occurred along branches connecting parasitic clades to their closest nonparasitic ancestors. Given that OBP represents a radical change in reproductive strategy, this shift may itself have contributed to lineage divergence, making divergence times informative about the environmental context of OBP evolution.

We mapped the paleotemperature (*31*) onto a recently published whole-genome phylogeny of 363 bird species (Bird 10000 Genomes Project) (*32*), which provided reliable divergence time estimates for four of the seven independent OBP origins (highlighted as red-shaded regions in Fig. 3B). For the remaining three origins, we consulted the

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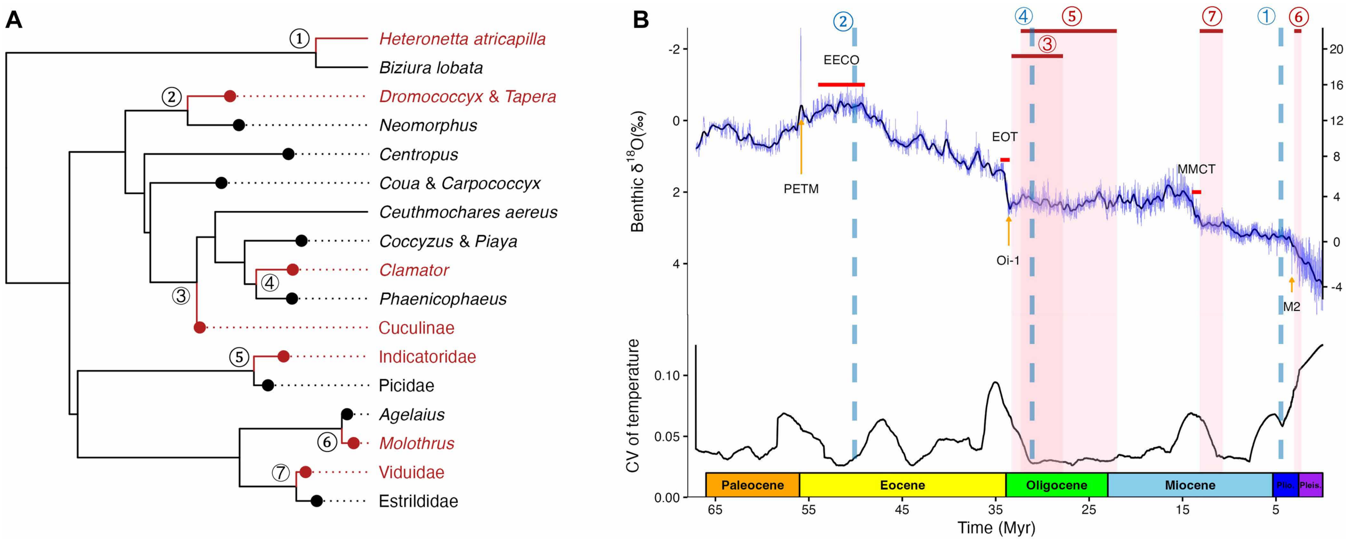

Fig. 3. Paleoclimatic conditions during the evolutionary origins of OBP. (**A**) Phylogenetic relationships between parasitic clades (red) and their closest parental care relatives (black). The seven independent evolutionary origins of OBP are marked by numbered labels (from one to seven). The phylogenetic tree in (A) was obtained from birdtree.org (*76*) for illustrative purposes. (**B**) Paleoclimate mapping. The upper panel in (B) shows the global $\delta^{18}O$ curve from Westerhold *et al.* (*31*), alongside estimates of the approximate mean surface temperature differences relative to present-day conditions. The lower panel in (B) shows the CV of temperature calculated with a ±2.5-Myr sliding window centered on each focal time point. Shaded red bars in (B) indicate the estimated divergence windows (with 95% confidence intervals) for the four OBP transitions dated from Stiller *et al.* (*32*). Blue dashed lines mark the approximate timing of the remaining three OBP origins based on the OneZoom dataset. EECO, Early Eocene climatic optimum.

OneZoom Tree of Life (*33*) to approximate divergence times from their closest nonparasitic relatives.

Temporal correspondence with climatic upheavals

Our analysis reveals a notable pattern: All four origins with reliable genomic divergence estimates occurred following major global cooling events associated with peaks in the temperature variability, as indicated by the coefficient of variation (CV) of paleotemperatures (Fig. 3B). Specifically, Cuculidae and honeyguides (Indicatoridae) diverged after the Eocene/Oligocene transition (EOT) and Oi-1 (the first major glacial period in the Oligocene) glaciation, Vidua finches (Viduidae) diverged after the Middle Miocene climate transition (MMCT), and cowbirds (*Molothrus*) diverged after the M2 (first major glacial event in the North Hemisphere) glaciation in the North Hemisphere. The remaining three OBP lineages show similar temporal associations with major environmental changes (see blue dashed lines in Fig. 3B): New World parasitic genera *Tapera* and *Dromococcyx* diverged from their parental care relatives after the Paleocene-Eocene thermal maximum (PETM), *Clamator* cuckoos after the EOT, and the parasitic waterfowl *Heteronetta atricapilla* during recent environmental fluctuation [<5 million years (Myr) ago] (*27*).

Mechanisms linking climate change to OBP evolution

Each of those above climatic events created conditions that likely elevated parental care costs, consistent with our modeling predictions. The PETM (~56 Myr) was a "hyperthermal" event characterized by rapid global warming of 5° to 7°C (*34*), causing widespread fauna displacement (*35*) that would have increased mortality and resource competition. The EOT marked a marked transition from "Warmhouse" to "Coolhouse" conditions (*31*), establishing of permanent ice sheets on Antarctica, and causing sea level fall (*36*, *37*). This transition accelerated speciation rates across multiple taxa [e.g., (*38*–*40*)], likely through habitat fragmentation and niche diversification (*41*). Notably, three of seven OBP origins also appear to have occurred after the EOT (Fig. 3B).

Similarly, the MMCT was characterized by pronounced global cooling and Antarctic ice expansion following the warm Miocene climate optimum (*31*), accompanied by substantial terrestrial ecosystem changes, including the expansion of grasslands and savannas at the expense of forests (*42*), particularly in Africa (*43*), where Vidua finches are found (*44*). The Middle Miocene therefore marked the first major transition of forest-adapted species into open, arid environments (*45*–*47*). Even the seemly warm Late Pliocene (3.6 to 2.58 Myr) exhibited notable climatic variability, including the intense M2 glaciation period (~3.312 to 3.264 Myr) (*48*), immediately preceding *Molothrus* cowbird origins (3.05 to 2.28 Myr) (*32*). In addition, the closure of the Panama Isthmus (~3 Myr ago) triggered the "Great American Biotic Interchange" (*49*), introducing previously unknown predators and competitors that disrupted native ecosystems (*50*, *51*). These changes likely could increase the costs of energetically demanding parental care behavior in birds, thereby promoting the evolution of OBP, as indicated by our theoretical model.

Timing patterns reveal a two-phase process

Notably, OBP lineage divergences do not coincide exactly with paleoclimate change events but rather mostly follow them (Fig. 3). To assess whether the origins of OBP follow major climatic transitions more often than expected by chance, we conducted binomial tests against randomized divergence timings (see table S1 for details). The results show that the probability of observing the divergence patterns under random timing ranges from 0.003 to 0.017 across different posttransition window definitions, indicating a significant tendency for OBP origins to occur after major climatic changes. This temporal offset suggests that the fixation of parasitism may not have taken place under peak environmental stress itself but rather during subsequent phases when ecological conditions were more stable or improving. In these posttransition phases, recovering ecosystems may have offered greater host availability and intensified competition as populations rebounded and previously suppressed species reexpanded into overlapping niches, factors that could favor parasitic individuals and promote the life-history modifications (e.g., increased fecundity, shifts to lower-quality habitats, or migration), which help maintain and reinforce the parasitic strategy.

Phylogenetic comparative analyses reveal life-history correlates of OBP

According to our modeling results, even if brood parasitism had initially evolved under elevated parental care costs and environmental stress, its persistence requires the evolution of additional life-history adaptations, particularly as conditions improve or stabilize. Previous analyses of cuckoo evolutionary pathways suggest that the parasitism is often followed by increases in migration and shifts to less productive habitats (*1*). We propose that these life-history transitions are not merely consequences of OBP evolution but rather key adaptations that actively drive the fixation of the parasitic strategy in populations by sustaining the cost-benefit dynamics identified in our models. Without these secondary adaptations, lineages that initially adopted parasitism might have reverted to parental care or gone extinct rather than persisting as obligate brood parasites (*15*, *22*).

To test these predictions comprehensively, we conducted phylogenetic comparative analysis examining the relationships between the reproductive strategy (parasitic versus nonparasitic) and key life-history traits across all OBP species with available data. Specifically, we regressed the reproductive strategy against body mass, trophic level, habitat type, habitat density, migration, environmental stress [first and second principal components (PC1 and PC2, respectively)], and absolute latitude (see Materials and Methods).

High-mortality life-history traits promote parasitism

Our results illustrate that parasitic species are significantly more likely to occur in open habitats [credible interval (CI) = 0.160 to 10.728; *P* value from Markov-chain Monte Carlo (pMCMC) = 0.038] and among full migratory birds (CI = 2.093 to 24.429; pMCMC = 0.002) than sedentary species or those inhabiting dense habitats (Fig. 4 and table S2). The open-habitat structure generally lacks the structural complexity that provides refuge from predators, exposing adults to higher predation risks (*52*), while migration can also impose additional survival costs by increasing energy expenditure and exposure to unpredictable conditions (*53*). Both traits elevate adult mortality (i.e., elevating the costs of parental care), matching the theoretical expectation that higher mortality shifts the cost-benefit balance away from direct parental care and toward parasitism (see Figs. 1 and 2).

Environmental harshness and latitude interact to shape parasitic evolution

We also examined the role of environmental stress using PCs derived from six bioclimatic variables. The PC1 explained 50.7% of the variance, with lower values indicating environments that are colder and drier during the breeding seasons and characterized by less predictable temperatures and greater temperature seasonality (table S3). In accordance with previous research (*54*), we refer to this axis as environmental harshness. Accordingly, we classify environments

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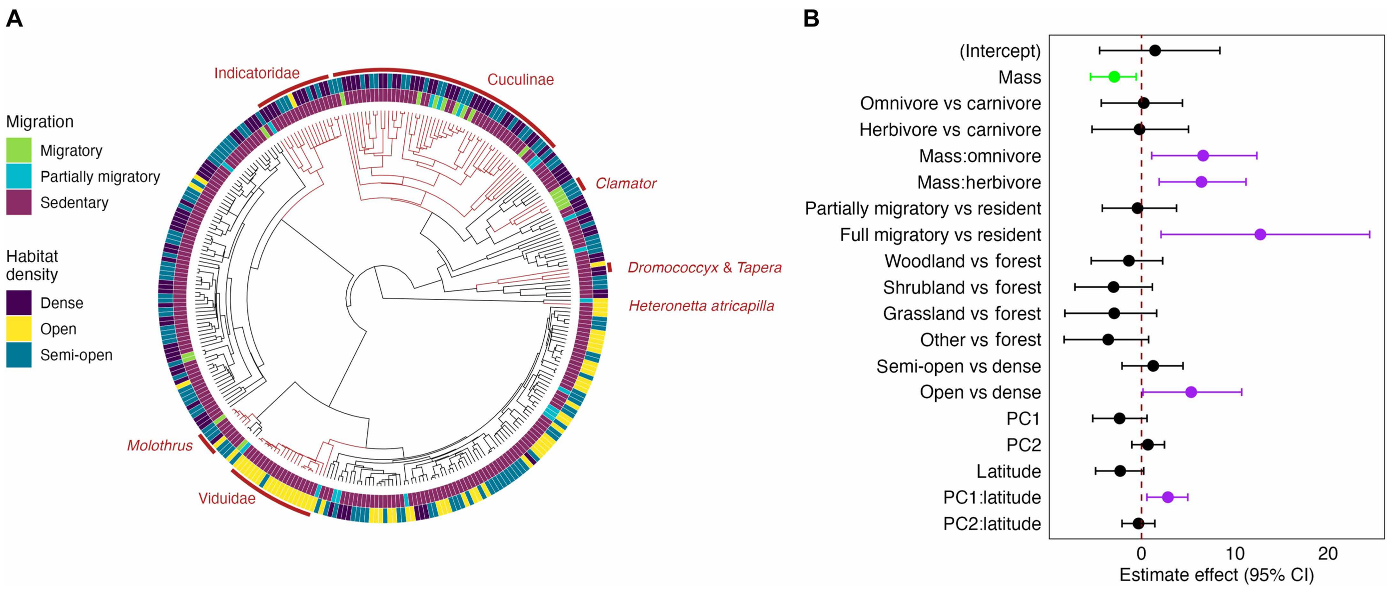

Fig. 4. Results of the phylogenetically controlled Bayesian phylogenetic mixed model predicting the breeding strategy (parasitic versus nonparasitic) in birds (*n* = 300). (**A**) Phylogenetic distribution of parasitic and nonparasitic breeding strategies, with the consensus tree for illustration. (**B**) Estimated coefficients of different predictor variables. Points indicate the mean effect size for all predictor variables and lines indicate 95% CIs. Predictors with significant (pMCMC < 0.05) negative and positive effects are shown in green and purple, respectively. For additional details, see table S2.

with low PC1 values as harsh as a result of their typically low productivity and unpredictable resource dynamics. Greater temperature seasonality in such environments also raises the likelihood of extreme climatic events, potentially elevating mortality and reducing reproductive success. In contrast, higher PC1 values correspond to warmer, wetter, and more predictable environments, which we consider benign. The PC2 explained 23.2% of the variance and reflected the variation in precipitation within and among years (table S3).

Parasitic propensity was significantly predicted by an interaction between PC1 and latitude (Fig. 4 and table S2: CI = 0.577 to 4.953; pMCMC = 0.004). At low latitude, parasitism was more likely under harsh environmental conditions (low PC1 values), consistent with our theoretical predictions that occupying poor habitats enables the equilibrium to shift toward parasitism (see the gray arrow in Fig. 1A). However, at high latitudes, parasitism was more likely under benign conditions instead (Fig. 5, A to C). This latitude-dependent pattern likely reflects ecological constraints unique to high-latitude environments. At extreme latitudes, environmental harshness may exceed the feasibility threshold for parasitism (see Fig. 5C), making OBP less viable. High-latitude environments impose constraints such as severely shortened breeding seasons (*55*), exceptionally where temperature variability is extreme (*56*). Moreover, potential host species may be relatively rare in such environments (*57*), further limiting opportunities for successful parasitism (an effect similar to elevating *ra* values as shown in our model). Therefore, under extremely harsh, high-latitude conditions, potential hosts may be too rare or breeding windows too brief for parasitic strategies to succeed, regardless of their theoretical advantages.

Notably, we found no correlation between parasitic propensity and precipitation variability variable (PC2) or its interaction with latitude (table S2), suggesting that variation in temperature-related environmental factors exerts stronger selective pressure on parasitic evolution than precipitation patterns. This asymmetry parallels previous findings that temperature variability, rather than precipitation, correlates with host phylogenetic diversity of avian brood parasites (*58*) and with rates of infidelity and divorce in birds (*59*). One possible explanation is that temperature likely has more direct physiological effects on incubation success, offspring development, and even adult survival in birds (*60*), thereby more strongly influencing the cost-benefit balance of parental care. In contrast, although precipitation influences food availability (*61*), its effects are often more temporal and localized or can be buffered in certain ecosystems. We suggest that a deeper exploration of how temperature variability shapes different avian life histories may yield insights into the ecological and evolutionary drivers of parasitic breeding systems.

Body size effects vary with trophic levels

Parasitic birds tend to be smaller in body mass (Fig. 4 and table S2: CI = −5.459 to −0.573; pMCMC = 0.008), but this relationship varies significantly across trophic levels (95% CI for the interaction effect between body mass and trophic levels did not span zero; see Fig. 4B). Among carnivores, smaller species were more likely to be parasitic (Fig. 5D), while in herbivorous and omnivorous species, larger body mass was positively associated with parasitic propensity (Fig. 5, E and F).

These contrasting patterns likely reflect the difference in resource distribution and competitive dynamics across trophic levels. Carnivorous birds in our dataset primarily forage on insects, which are typically nutrient-rich and high-quality food resources (*62*). Such resources are often patchily distributed (*63*) and thus can be economically defensible. Larger carnivores may monopolize insect-rich patches, forcing smaller competitors to experience disproportionately higher survival and parental care costs, particularly under environmental stress. Our theoretical framework predicts that this

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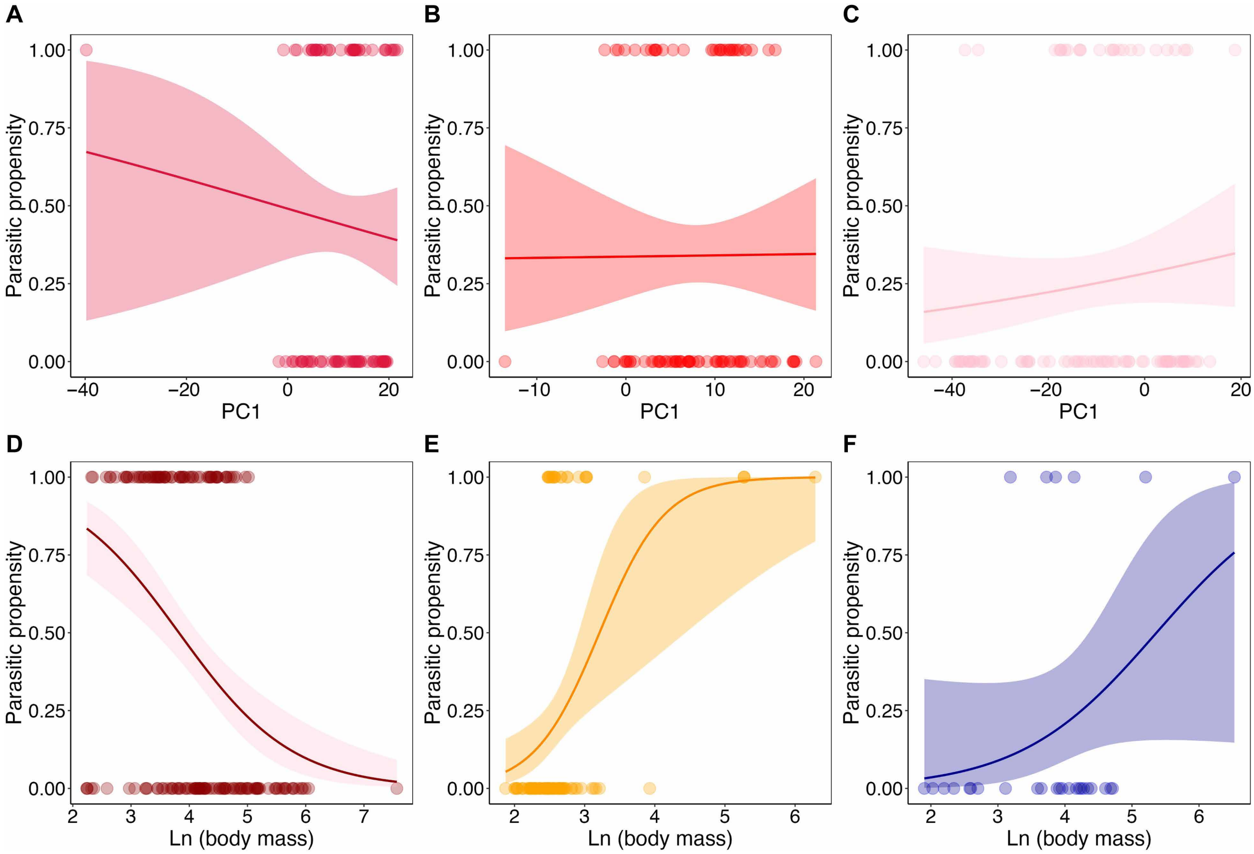

Fig. 5. Significant interaction effects on parasitic propensity. (**A** to **C**) Interaction effect between latitude and the environmental harshness variable (PC1); (**D** to **F**) interaction effect between trophic level and body mass. For illustration, we arbitrarily divided the latitude into three categories of low, medium, and large using equal quantiles. The results for these three types of latitude are given in (A) to (C), respectively. The results for carnivorous, herbivorous, and omnivorous birds are given in (D) to (F), respectively. See table S2 for additional details. Trendlines represent logistic regressions with 95% confidence intervals (shaded bands) plotted for illustrative purpose.

combination of higher parental care costs and elevated mortality should favor parasitic strategies. In addition, the trade-offs associated with body size can be complex and context-dependent (*64*). For instance, compared to larger individuals, smaller ones may exhibit lower nutrient conversion efficiency because of their higher mass-specific metabolic rates (*65*), exploit fewer resource types (*66*), incur greater mass-specific energy costs of locomotion (*67*), and be less capable of buffering against resource fluctuations (*68*). These multiple factors may play a combined role in driving the small carnivorous birds to have a higher parasitic propensity, which deserves much more empirical and theoretical exploration in the future.

Conversely, herbivorous and omnivorous birds (primarily seed eating in our dataset) rely on widely dispersed, low-quality resources like grass seeds. Larger individuals in these groups often face greater challenges in meeting their elevated energetic demands, particularly under poor environmental conditions, potentially reducing their fitness or parental efficiency (*69*). In our theoretical framework, this can translate to higher effective parental care costs for larger individuals in these trophic groups, again predicting the observed shift toward parasitic strategies. The potential energetic stress thus could potentially make parasitism more advantageous for larger herbivores and omnivores. Supporting this interpretation, recent studies show that larger herbivores currently face the highest extinction risk among different animal taxa (*70*), possibly reflecting their increased ecological vulnerability.

Additional factors may also influence size-parasitism relationships. For example, Bergmann's rule suggests that thermoregulatory demands drive larger body sizes in colder climates, potentially confounding relationships between size and parasitic evolution. Future research should explore how competitive ability, thermoregulation, and environmental variability interact to shape OBP evolution.

Environmental transitions drive OBP strategy fixation

To examine the ecological context of OBP strategy fixation, we also reconstructed ancestral environmental conditions (PC1) across OBP phylogenies (fig. S3). In most parasitic clades, PC1 values increased from the closest nonparasitic ancestors to the most recent common ancestor of parasitic lineages, suggesting that the fixation of the OBP strategy was often associated with shifts into more benign environments. This supports our earlier interpretation that parasitism may initially emerge under harsh conditions, but subsequent fixation and eventual speciation occurred more likely when environments improve and host availability increases.

However, Old World cuckoos, particularly *Clamator* species, showed an opposite pattern, with OBP fixation associated with shifts toward harsher environments (fig. S3B). This suggests that parasitic

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fixation may sometimes result from competitive exclusion from preferred habitats rather than ecological opportunity in improving environments. These contrasting patterns reveal the complexity of parasitic evolution and highlight the need for further work on how local ecological conditions can mediate the success and persistence of parasitic strategies across species.

DISCUSSION

Macroevolutionary framework for understanding OBP evolution

Discussions of brood parasitism have traditionally focused on coevolution dynamics between parasites and hosts, emphasizing proximate mechanisms underlying interspecific variation in parasitic and antiparasitic strategies (*6*, *9*). While this approach has yielded valuable insights into the maintenance of parasitism, it has limited our understanding of ultimate evolutionary drivers of OBP within a broader macroevolutionary context. Previous evolutionary path analyses provided important insights into OBP evolutionary history in cuckoos (*1*, *22*), but the fundamental question of why parasitism evolves initially has remained largely unresolved.

Our integrated approach, which combines theoretical modeling, paleoclimatic analysis, and phylogenetic comparative methods, addresses this gap by identifying the environmental and evolutionary drivers underlying the origins of OBP. Theoretical results demonstrate that elevated parental care costs, particularly during marked environmental transition, create conditions where the parasitic strategy becomes selectively advantageous. Paleoclimatic evidence supports these predictions: All OBP origins with reliable genomic divergence estimates occurred following major global climatic upheavals (Fig. 3), suggesting evolutionary hysteresis (*71*) where shifts to parasitism depend critically on environmental history. Although this apparent temporal association should be interpreted cautiously given the considerable uncertainty in molecular clock estimates (often spanning several million years) and the indirect nature of paleoclimatic proxies, convergent evidence from multiple independent approaches offers a valuable complement to traditional evolutionary studies of brood parasitism and provides a macroevolutionary framework for future testing as methods and data continue to advance.

Two-phase evolutionary model

Our findings support a two-phase model of OBP evolution. Initially, marked environmental changes trigger behavioral shifts to parasitism by elevating parental care costs beyond sustainable thresholds. Subsequently, the fixation and even diversification of OBP likely occurred during periods of environmental stabilization or improvement when recovering ecosystems provide greater host availability and ecological opportunity. This temporal decoupling between the behavioral origin and fixation may explain why OBP divergences from nonparasitic relatives often follow rather than coincide exactly with major paleoclimatic events (see Fig. 3).

The second phase requires evolution of specific life-history adaptations that maintain parasitic advantages even as conditions improve. Our comparative analyses reveal that successful parasitic lineages consistently exhibit high-mortality traits such as migration and open-habitat use, which continue to elevate parental care costs and reinforce the benefits of parasitism. Without these secondary adaptations, initially, parasitic lineages might revert to parental care or face extinction as environmental pressures subside.

Explaining convergent evolution across distant lineages

This framework illuminates why OBP has evolved convergently across phylogenetically distant orders with vastly different life histories, from waterfowl (Anseriformes) to honeyguides (Piciformes). Rather than requiring specific genetic or morphological prerequisites, OBP evolution depends primarily on ecological conditions (environmental stress) and life-history flexibility (ability to adopt high-mortality strategies). Given that these requirements are not phylogenetically constrained, parasitism can potentially emerge in any lineage experiencing appropriate ecological pressures and having sufficient behavioral plasticity.

Our finding that both smaller carnivorous and larger herbivorous/omnivorous species show elevated parasitic propensity further supports this ecological framework. The contrasting body size relationships across trophic levels reflect different resource distribution patterns and competitive dynamics rather than phylogenetic constraints, demonstrating that multiple evolutionary pathways can lead to successful parasitism.

Constraints on parasitic persistence

Despite its apparent macroevolutionary potential, OBP remains remarkably rare (only ~1% in birds), indicating substantial constraints on parasitic establishment and persistence. Several factors likely contribute to this rarity beyond the environmental triggers we identified. First, imprinting and species recognition present fundamental challenges for parasitic offspring raised by heterospecific hosts, potentially interfering with mate choice and reproductive success in subsequent generations (*72*). Second, the difficulty in locating and successfully exploiting suitable hosts, particularly under fluctuating ecological conditions, creates ongoing selective pressure against the establishment of parasitism.

In addition, the current phylogenetic record represents only successful parasitic lineages that survived complex evolutionary transition. Many parasitic experiments might have failed during coevolutionary arms races with hosts or because of demographic stochasticity, leaving no trace in the contemporary record. As such, the seven surviving OBP lineages may thus represent a small fraction of attempted parasitic transitions throughout avian evolutionary history, with multiple biological, ecological, and demographic filters determining which lineages persist.

Model limitations and future directions

While our model emphasizes adult mortality associated with parental care under environmental stress, we acknowledge that this may not fully capture the full breadth of selective pressures involved. The Hamilton and Orians hypothesis (*19*) suggests that high nest predation rates may directly promote parasitic evolution by making alternative reproductive strategies advantageous after nest failure. This mechanism has received empirical support from both field studies (*73*, *74*) and controlled experiments (*75*). We suggest that integrating clutch predation, such as via a separate term or risk function, would be a valuable extension in future models.

Another important consideration is differential environmental sensitivity between species. When potential host species are less vulnerable to environmental stress than would-be parasites, offspring survival is higher for parasitic strategies. This survival asymmetry can effectively lower the threshold conditions for OBP evolution (see fig. S2). Climate change may further amplify these effects by indirectly influencing nest predation through alterations in vegetation

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structure or predator activity patterns, creating additional pathways linking environmental variability to parasitic evolution. Future theoretical and empirical work should explore these connections and test our model's predictions about the roles of increased fecundity and longevity in reinforcing parasitic advantages.

Synthesis and implications

Our findings suggest that OBP evolution may arise from a complex interplay between environmental triggers, life-history constraints, and ecological opportunities. Theoretical results indicate that elevated parental care costs can favor the emergence of parasitic strategies, while periods of environmental instability may have increased these costs and reduced offspring survival, creating conditions under which brood parasitism could become selectively advantageous. The subsequent evolution of high-mortality life-history traits then enables long-term persistence and establishment of OBP lineages. This framework provides the first comprehensive explanation for the repeated yet rare evolution of OBP, linking environmental pressures to evolutionary outcomes across multiple temporal and taxonomic scales.

Understanding these evolutionary drivers has profound implications for predicting how reproductive strategies might respond to ongoing environmental changes, as well as for identifying the ecological conditions that favor alternative reproductive strategies across taxa. As global environmental change accelerates, our framework suggests that species facing elevated parental care costs may be candidates for evolutionary transitions toward alternative reproductive strategies (i.e., parasitism), although the rarity of successful transitions emphasizes the stringent conditions required for such evolutionary innovations.

MATERIALS AND METHODS

Optimization model

We consider a population of individuals that adopt a reproductive strategy defined by the parasitism rate p, representing the probability that an individual chooses to parasitize rather than raise offspring itself during a reproductive bout. Without the loss of generality, we assume that reproduction occurs in discrete bouts (generations), with each individual reproducing once per bout (*23*).

When an individual opts to parasitize in a given bout, it faces a probability a of detection by the host. Upon detection, the host may reject (either the egg or chicken) or abandon the parasitized nest with probability r. Thus, the probability of a successful parasitic attempt is $a(1-r)+1-a$ (i.e., $1-ar$), accounting for either nondetection or acceptance after detection. Furthermore, we assume that individuals that parasitize or care for their own offspring produce f_1 and f_2 offspring, respectively.

Using these above definitions, we have the expected reproductive fitness φ of an individual with parasitism rate p during one bout

$$\varphi = p(1-ar)f_1 + (1-p)f_2$$

where the first term [i.e., $p(1-ar)f_1$] corresponds to expected successful offspring via parasitism and the second term [i.e., $(1-p)f_2$] via parental care.

Between reproductive bouts, individuals experience an environment-dependent baseline mortality rate d, which varies with environmental condition $x \in (0,1)$. We explore three different scenarios for the function $d(x)$, which are shown in Fig. 2. The linear function of $d(x) = 1-x$ represents a uniform decline in mortality as conditions improve, which can be used as a control state for comparisons. The concave function (modeled using $d(x) = \frac{1-x}{mx+1-x}$, where $m > 1$ determines the steepness of the line) models a scenario when individuals are tolerant to environmental deterioration where mortality increases slowly at first and then accelerates. The convex lines [modeled using $d(x) = 1-x^{\alpha}$, where $\alpha > 1$] model a scenario when individuals are sensitive to environmental change where mortality rises sharply initially and then slows.

As providing care to offspring is often costly (*3*), we assume that parental care imposes an additional mortality cost c per bout. In this case, the probability that an individual with parasitism rate p survives to the next bout is

$$s = 1 - \left[d + (1-p)c\right]$$

We assume that the additional death rate c is also environmentally dependent. For simplicity, we will use $c = kd$ to model the parental care costs, where $k > 0$. To guarantee that the survival probability s is larger than zero for any value of p, $k < \frac{1}{d} - 1$ is required.

Following Lerch and Servedio (*23*), we can then derive the expected lifetime reproductive fitness F of an individual with parasitic rate p, that is

$$F = \sum_{t=1}^{\infty} \varphi s^t$$

which converges to $\frac{s}{1-s}\varphi$. Maximizing fitness involves finding the value p^* such that

$$\frac{dF}{dp} = 0 \text{ and } \frac{d^2F}{dp^2} < 0$$

The sign of the fitness gradient $\frac{dF}{dp}$ at the boundaries indicates the evolutionary direction. Specifically, whenever $\frac{dF}{dp}\big|_{p=0} < 0$, parasitism is selected against when parental care is fixed. Similarly, whenever $\frac{dF}{dp}\big|_{p=1} > 0$, parental care is selected against when parasitism is fixed.

Paleoclimate and independent evolutionary origins of OBP in birds

To illustrate the paleoclimatic context in which OBP lineages diverged from their parental care relatives, we first collected recently published benthic $\delta^{18}O$ (‰) measurements for the Cenozoic and converted these values into estimates of global mean surface air temperature (°C) using the formulas provided by Westerhold *et al.* (*31*). Furthermore, to assess whether long-term climatic instability may have contributed to the emergence of OBP, we calculated the CV of the paleotemperature dataset using a ±2.5-Myr sliding window centered on each focal time point.

Note that relatively accurate divergence timing is crucial for mapping OBP origins onto paleoclimatic trends. We therefore used a recently published time-calibrated avian phylogeny based on whole-genome data from 363 species representing 218 bird families (*32*), which provides divergence time estimates with 95% confidence intervals for each node. From this phylogeny, we identified four independent evolutionary origins of OBP (Fig. 3). For the remaining three origins not covered by the genomic dataset, we referred to the OneZoom Tree of Life (*33*), which incorporates the comprehensive birdtree.org phylogeny (*76*), to obtain approximate divergence times from their closest nonparasitic relatives.

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Species reproductive strategy data collection

To evaluate the macroevolutionary patterns of avian brood parasitism and test predictions from our theoretical optimization model, we also conducted phylogenetic comparative analyses. We first compiled avian breeding system data from the published review literature (*77*), classifying each species as either nonparasitic (i.e., parental care) or parasitic. Parasitic species are defined as those in which parents never raise their own offspring (i.e., OBP). As we primarily attempt to assess the evolution and maintenance of OBP in birds, and it has been acknowledged that OBP is more likely to evolve directly from parental care (*2*), we therefore restricted our comparison to species that engage in parental care (i.e., either male only, female only, or paired) as the nonparasitic group.

OBP has independently evolved three times among order Cuculiformes (cuckoos), which contains the greatest number of parasitic species (65 species). We therefore included all cuckoo species with available data. OBP has also evolved twice among order Passeriformes (once in genus *Molothrus* in family Icteridae and once in family Viduidae), once among Piciformes (in family Indicatoridae), and once among Anseriformes (only in black-headed duck *H. atricapilla*). To strengthen the statistical power and minimize confounding from distantly related taxa [which may differ in many traits unrelated to the reproductive strategy; see ref. (*78*)], we included closely related nonparasitic lineages for comparison. Specifically, we added genus *Agelaius* (sister to *Molothrus*), family Estrildidae (sister to Viduidae), family Picidae (sister to Indicatoridae), and species *Biziura lobata* (sister to *H. atricapilla*). In total, we categorized 103 species as parasites (~94% of all known OBP species in birds) and 197 species as nonparasites.

Compiling data on life history traits and environmental stress

We obtained species-level life history and ecological trait data from AVONET, a recently published global dataset that provides comprehensive functional trait information for all bird species (*79*). Specifically, the traits we extracted include the following: adult body mass, migration status (1 for sedentary, 2 for partially migratory, and 3 for migratory), habitat density (1 for dense habitats, 2 for semi-open habitats, and 3 for open habitats), trophic level (carnivore, herbivore, or omnivore), and centroid latitude of distribution. Habitat density was used as a proxy for environmental conditions, with open habitats generally assumed to be more resource-poor and subject to higher predation risk (*80*, *81*). As another important life-history trait, migration status was also included, given that migratory behavior may increase mortality as a result of high energetic costs (*82*). Habitat type was included as a potential confounding variable. The original AVONET dataset defines nine habitat categories (*79*). We merged those categories that contained fewer than five parasitic species, resulting in five broad categories (i.e., forest, grassland, shrubland, woodland, and other).

In addition, to assess environmental stress in breeding habitats between parasitic and nonparasitic species, we downloaded geographic range maps for all species from BirdLife International (*83*) and extracted the environmental variables within their breeding ranges. Specifically, we used (i) the precipitation and mean temperature of the wettest quarter, representing the productivity of the breeding area (*84*); (ii) temperature seasonality (standard deviation × 100) and precipitation seasonality (coefficient of variation), representing the degree of change in climate; and (3) Colwell's predictability indices (*85*) for temperature and precipitation (*58*). These environmental variables were downloaded from two resources: the WorldClim bioclimatic dataset at a 5-min spatial resolution (~10 km^2) (*86*) and global gridded temperature and precipitation predictability data at a 30-s resolution (~1 km^2) from Jiang *et al.* (*87*). To ensure consistency across datasets, the predictability layers were first resampled to match the resolution of WorldClim data. We then calculated the mean values of all environmental variables across the entire breeding range of each species. Before the comparative analysis, we conducted a phylogenetic PC analysis to reduce the dimensionality of environmental variables (six), implemented using the "phyl.pca" function from the phytools R package (*88*). We selected the PC1 and PC2 that had eigenvalues greater than one for further analysis. We reported the PC loadings and percentage of variance explained for each PC in table S3.

Phylogenetic trees

We randomly downloaded a sample of 1000 full phylogenetic trees from BirdTree (*76*) with the Hackett backbone. We subsequently trimmed them to the species for which we had breeding strategy and other life-history data. Using these 1000 trees, we generated a consensus tree for further analyses, which was implemented using the "ls.consensus" function with the least-square method in the phytools R package (*88*).

Bayesian comparative analysis

To explore how different life-history traits and environmental stress are related to parasitic propensity, we fitted the following model: The reproductive strategy (parasitic or nonparasitic) was set as a binary response variable and log-transformed body mass, habitat type, habitat density, migration status, trophic level, two environmental PCs (PC1 and PC2), and the absolute central latitude as predictors. We also included two key interaction terms to capture potential context-dependent effects. First, the interaction between body mass and trophic level was used to test whether body size effects on OBP evolution depend on the dietary strategy, as energetic constraints and foraging demands can vary substantially across trophic niches and body sizes (*89*). Second, the interactions between latitude and PC1/PC2 were included to examine whether environmental stress effects vary across latitudinal gradients, given that relatively high-latitude species often experience greater climatic seasonality and unpredictability (*90*), potentially shaping their responses to local climatic conditions. All continuous explanatory variables were mean centered and scaled to have an SD of 1. We calculated the variance inflation factors for all variables included in the model and found that all values were below 5 (see table S2), indicating low multicollinearity.

We used a Bayesian approach implemented in the MCMCglmm R package (*91*) with weakly informative priors ($V = 1$, $\nu = 1000$, $\alpha.\mu = 0$, and $\alpha.V = 1$) (*92*). For the binary response variable, the residual variance cannot be estimated, so we fixed it to 1. We ran for 8,005,000 iterations with a 5000 burn-in and a thinning of 2000 iterations, ensuring that the effective sample sizes were all above 1000. Using Gelman and Rubin's convergence diagnostic test, we also confirmed the model convergence (with potential scale reduction factor <1.1). We used the consensus tree as a random effect and reported the estimates and pMCMC values.

To account for phylogenetic uncertainty, we also ran the model on 1000 randomly selected phylogenetic trees with the Hackett

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backbone and 1000 randomly selected trees with the Ericson backbone also downloaded from birdtree.org (*76*). Each mcmcglmm chain was run for 105,000 iterations, with a burn-in of 5000 and a thinning interval of 2000. The resulting outputs of each dataset were then combined into a single pseudoposterior distribution and analyzed accordingly. As the results were qualitatively similar to those analyzed using the consensus tree, i.e., supporting the same conclusions (table S4), we therefore present the results with the consensus tree in the main text.

Reconstruction of ancestral environmental characters

Last, to reconstruct the ancestral values of the environmental variable PC1, we used the function "fastAnc" from the phytools R package (*88*) to estimate PC1 values at each internal node of the phylogeny. For each OBP clade, we then calculated the change in PC1 from the most recent divergence between the OBP lineage and its closest nonparasitic (parental care) lineage to the most recent common ancestor of the OBP clade. To account for the phylogenetic uncertainty, these estimates were repeated across 1000 tree topologies with the Hackett backbone. For visualization, we also used the function "contMap" in the phytools R package to plot the PC1 values onto the consensus tree (see fig. S3A).

Supplementary Materials

This PDF file includes:
Figs. S1 to S3
Tables S1 to S4

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